

Lab Task 1: Bank Account Management System (Basic):

```
Microsoft Visual Studio Debu X + ~ - □ X

===== Account Information =====
Bank Name: Global Trust Bank
Account Name: Suman Shrestha
Account Number: 123456
Account Type: Generic
Balance: 2222.0
Active: True

Transaction History:
- Account created for Suman Shrestha. Account Number: 123456
- Deposited: 2345
- Withdrawn: 123

C:\Users\User\source\repos\LabWork\Dotnet\Lab-1And2\bin\Debug\Lab-1And2.exe (process 11724) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

Lab Task 2: Bank Account Management System with Inheritance and Polymorphism

```
Microsoft Visual Studio Debu X + - X
Bank Name: Global Trust Bank
Account Name: Suman Shrestha
Account Number: 123456
Account Type: Generic
Balance: 2222.0
Active: True

Transaction History:
- Account created for Suman Shrestha. Account Number: 123456
- Deposited: 2345
- Withdrawn: 123
===== Account Information =====
Bank Name: Global Trust Bank
Account Name: Suman Shrestha
Account Number: 123456
Account Type: Generic
Balance: 197515.76
Active: True

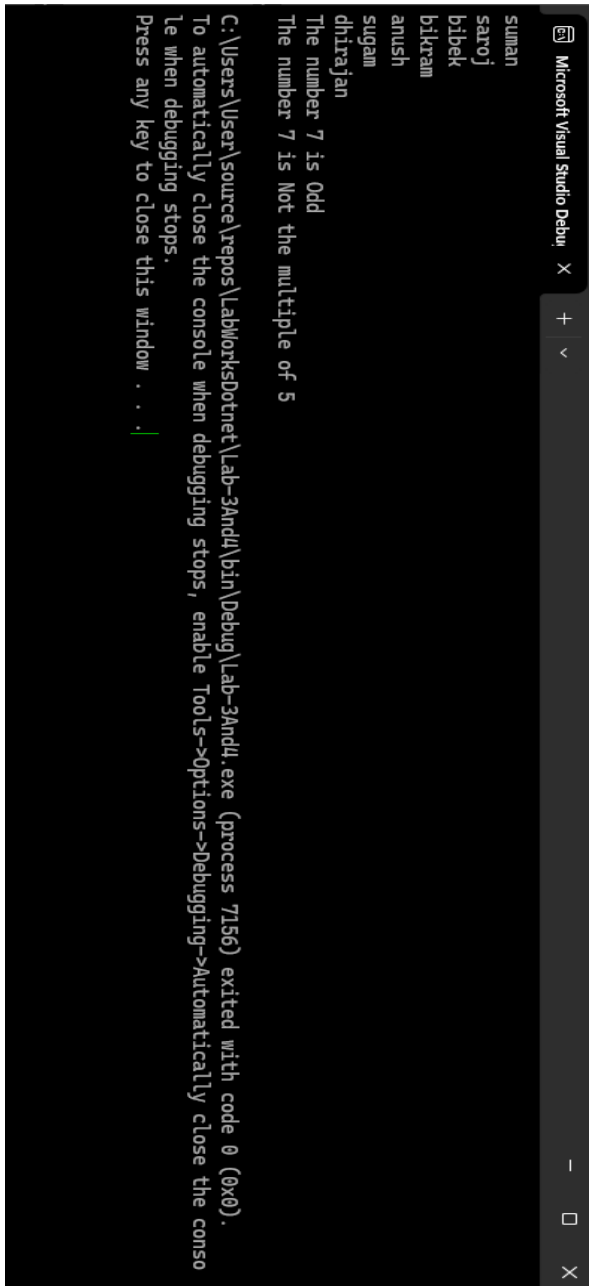
Transaction History:
- Account created for Suman Shrestha. Account Number: 123456
- Deposited: 200020
- Withdrawn: 10101
- Deposited: 7596.76
Interest Rate: 4%

C:\Users\User\source\repos\labwork\Dotnet\Lab-1And2\bin\Debug\Lab-1And2.exe (process 15340) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . . |
```

Lab Task 3: Understanding Single-cast and Multi-cast Delegates

```
Microsoft Visual Studio Debu X + - X
Single-cast Delegate:
Welcome to the program!
Multi-cast Delegate:
Welcome to the program!
Goodbye! Have a nice day!
After Removing ShowWelcome():
Goodbye! Have a nice day!
C:\Users\User\source\repos\LabWork\Dotnet\Lab-3\ndd\bin\Debug\Lab-3\ndd.exe (process 24424) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

Lab Task 4: Using Lambda Expressions with Action, Func, and Predicate



```
Microsoft Visual Studio Debug Console
suman
saroj
bibek
bikram
anush
sugam
dhiraajan
The number 7 is Odd
The number 7 is Not the multiple of 5
C:\Users\User\source\repos\LabWork\dotnet\Lab-3And4\bin\Debug\Lab-3And4.exe (process 7156) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console
when debugging stops.
Press any key to close this window . . .
```

Lab Task 5 :Working with LINQ in C# using Arrays and Collections

```
All The Active Employees:
Ram Sharma
Sita Thapa
Hari Karki
Bikash Adhikari
Anita KC
Prakash Rai
Mina Shrestha

Sorted by descending order of salary:
Bikash Adhikari - 90000
Mina Shrestha - 85000
Hari Karki - 75000
Suman Gurung - 70000
Dipak Bista - 65000
Sita Thapa - 60000
Anita KC - 55000
Gita Poudel - 50000
Ram Sharma - 45000
Prakash Rai - 40000

All the Departments:
IT
HR
Finance
Marketing
Sales
False
Ram Sharma - 45000
Sita Thapa - 60000
Hari Karki - 75000
Gita Poudel - 50000
Bikash Adhikari - 90000
Anita KC - 55000
Suman Gurung - 70000
Prakash Rai - 40000
Mina Shrestha - 85000
Dipak Bista - 65000

C:\Users\User\source\repos\LabWorkDotnet\Lab-Sand\bin\Debug\Lab-Sand.exe (process 6244) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

Lab Task 6: C# File I/O – Note Management System

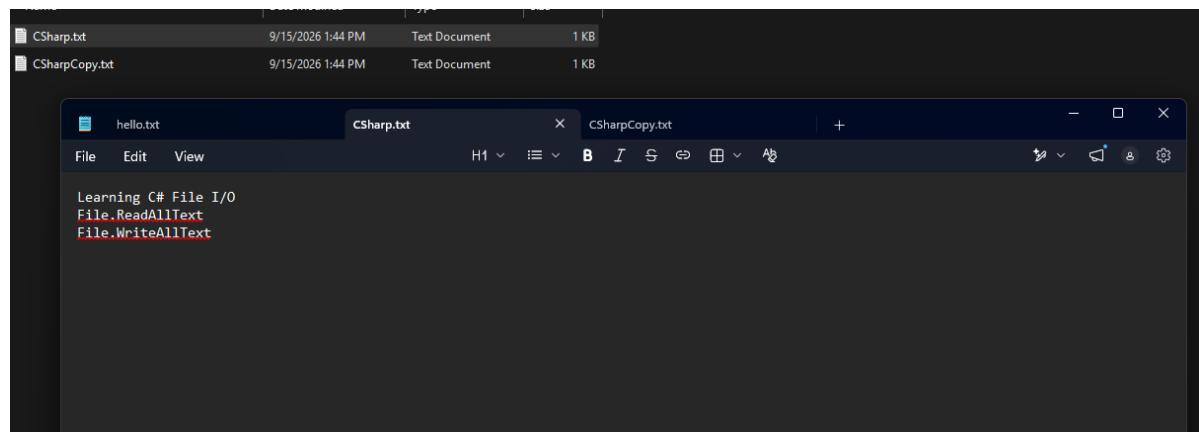
```
Microsoft Visual Studio Debug Console
NotesApp directory created.
Note created successfully.

--- Note Content ---
Learning C# File I/O
File.ReadAllText
File.WriteAllText

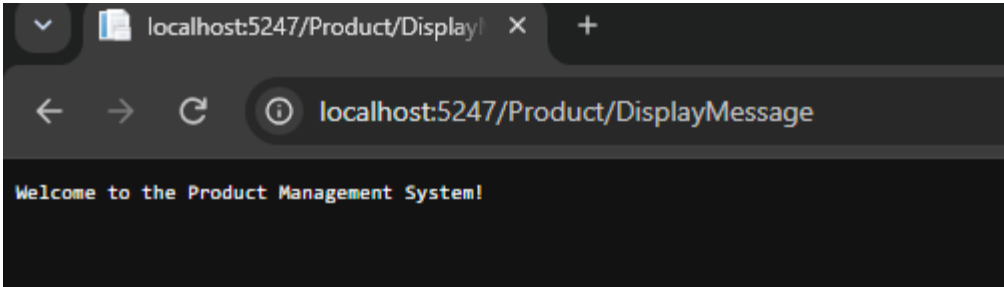
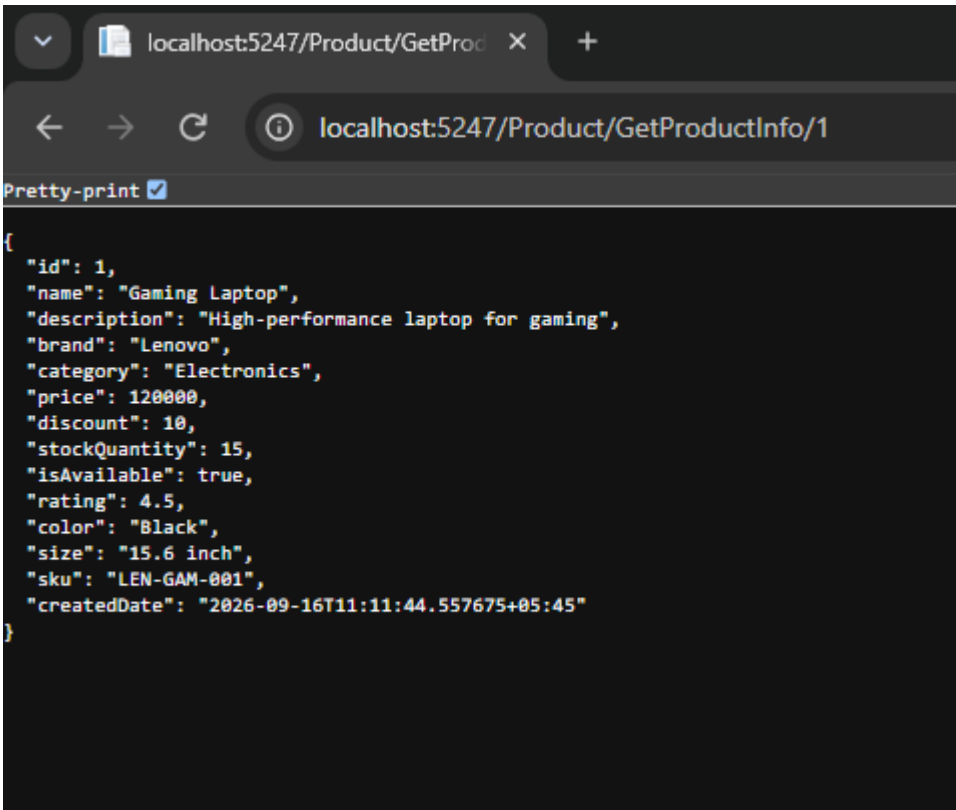
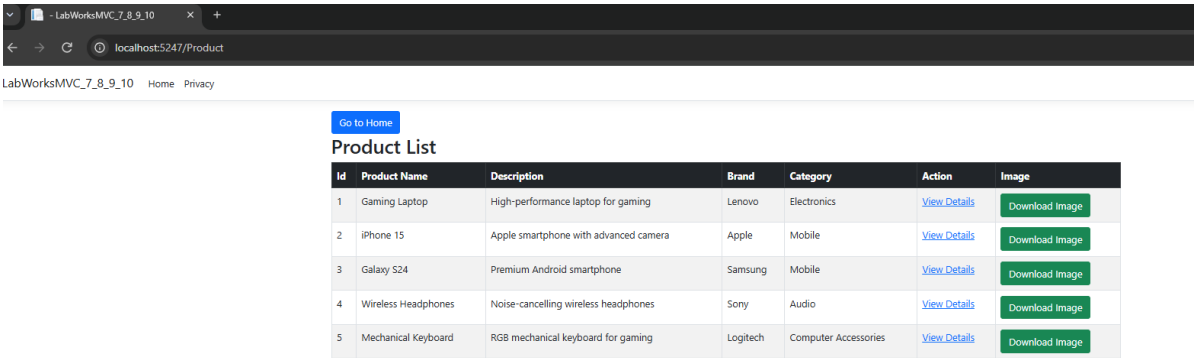
--- All Notes ---
CSharp
Note copied successfully.
Note renamed successfully.

--- Note Line By Line ---
1: Learning C# File I/O
2: File.ReadAllText
3: File.WriteAllText
Are you sure you want to delete 'CSharpCopy'? (y/n): y
Note deleted successfully.

C:\Users\User\source\repos\LabWorksDotnet\Lab-5And6\bin\Debug\Lab-5And6.exe (process 2368) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```



Lab Work 8: Understanding Controllers and ActionResult in ASP.NET Core



Lab Work 9: Razor Syntax, Tag Helpers, and Custom Tag Helpers in ASP.NET Core MVC



Welcome to Razor Demo!

Numbers:

- 10 - Number is small
- 20 - Number is large
- 30 - Number is large
- 40 - Number is large
- 50 - Number is large



Custom Tag Helper

Current Date: 09/17/2026

User Form

Name		Name is required
Email		Email is required
<input type="submit" value="Submit"/>		

Lab Work 10: Model Binding & Validations in ASP.NET Core MVC

Create Person - LabWorkSMC

localhost:5277/ModelBinding/Index

LabWorkSMC_7.8.9.10 Home Privacy

Create a Person

Name

Email

test1

Name is required

Age

2

Please enter a valid email address

Age must be between 18 and 100

Submit

Lab Task 11: Working with ADO.NET using SqlCommand and SqlDataReader

```
Microsoft Visual Studio Debug Console
Active Employees:
-----
Name: Alice Johnson, Salary: $60,000.00
Name: Bob Smith, Salary: $55,000.00
Name: David Lee, Salary: $48,000.00

C:\Users\User\source\repos\LabWorksDotnet\LabWorks_11_12_13\bin\Debug\LabWorks_11_12_13.exe (process 13732) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

100 % No issues found

Results Messages

	EmployeeId	Name	Salary	Department	IsResigned	Experience
1	1	Alice Johnson	60000.00	IT	0	2
2	2	Bob Smith	55000.00	HR	0	5
3	3	Carol Davis	70000.00	Finance	1	8
4	4	David Lee	48000.00	IT	0	1
5	6	John Smith	55000.00	IT	0	3

Microsoft Visual Studio Debug Console

Employee inserted successfully.

C:\Users\User\source\repos\LabWorksDotnet\LabWorks_11_12_13\bin\Debug\LabWorks_11_12_13.exe (process 11772) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```
Microsoft Visual Studio Debug Console
Employee Data:
-----
Name: Alice Johnson, Department: IT, Experience: 2
Name: Bob Smith, Department: HR, Experience: 5
Name: Carol Davis, Department: Finance, Experience: 8
Name: David Lee, Department: IT, Experience: 1
Name: John Smith, Department: IT, Experience: 3

C:\Users\User\source\repos\LabWorksDotnet\LabWorks_11_12_13\bin\Debug\LabWorks_11_12_13.exe (process 23768) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .|
```

100 % 1 0

Results Messages

	EmployeeId	Name	Salary	Department	IsResigned	Experience
1	1	Alice Johnson	60000.00	IT	0	2
2	2	Bob Smith	55000.00	HR	0	5
3	3	Carol Davis	70000.00	Finance	1	8
4	4	David Lee	48000.00	IT	0	1
5	6	John Smith	55000.00	IT	0	4

Microsoft Visual Studio Debug Console

```
1 row(s) updated for 'John Smith'.

C:\Users\User\source\repos\LabWorksDotnet\LabWorks_11_12_13\bin\Debug\LabWorks_11_12_13.exe (process 9124) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .|
```

Lab Task 12: Working with Entity Framework using DbContext

Microsoft Visual Studio Debug Console

```
Active Employees:
-----
Name: Alice Johnson, Salary: $60,000.00
Name: Bob Smith, Salary: $55,000.00
Name: David Lee, Salary: $48,000.00
Name: John Smith, Salary: $55,000.00

C:\Users\User\source\repos\LabWorksDotnet\LabWorks_11_12_13\bin\Debug\LabWorks_11_12_13.exe (process 28784) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

Results Messages

	EmployeeId	Name	Salary	Department	IsResigned	Experience
1	1	Alice Johnson	60000.00	IT	0	2
2	2	Bob Smith	55000.00	HR	0	5
3	3	Carol Davis	70000.00	Finance	1	8
4	4	David Lee	48000.00	IT	0	1
5	6	John Smith	55000.00	IT	0	4
6	7	Emma Watson	62000.00	Marketing	0	4

Microsoft Visual Studio Debug Console

```
Active Employees:
-----
Name: Alice Johnson, Salary: $60,000.00
Name: Bob Smith, Salary: $55,000.00
Name: David Lee, Salary: $48,000.00
Name: John Smith, Salary: $55,000.00
Inserted employee with new Id: 7

C:\Users\User\source\repos\LabWorksDotnet\LabWorks_11_12_13\bin\Debug\LabWorks_11_12_13.exe (process 17084) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

ResultsMessages

	EmployeeId	Name	Salary	Department	IsResigned	Experience
1	1	Alice Johnson	60000.00	IT	0	2
2	2	Bob Smith	55000.00	HR	0	5
3	3	Carol Davis	70000.00	Finance	1	8
4	4	David Lee	48000.00	IT	0	1
5	6	John Smith	55000.00	IT	0	4
6	7	Emma Watson	62000.00	Marketing	0	5

Microsoft Visual Studio Debug Console

Emma Watson's experience updated to 5.

C:\Users\User\source\repos\LabWorksDotnet\LabWorks_11_12_13\bin\Debug\LabWorks_11_12_13.exe (process 432) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

ResultsMessages

1	2	Bob Smith	55000.00	HR	0	5
2	3	Carol Davis	70000.00	Finance	1	8
3	4	David Lee	48000.00	IT	0	1
4	6	John Smith	55000.00	IT	0	4
5	7	Emma Watson	62000.00	Marketing	0	5

Microsoft Visual Studio Debug Console

Employee with Id 1 deleted.

C:\Users\User\source\repos\LabWorksDotnet\LabWorks_11_12_13\bin\Debug\LabWorks_11_12_13.exe (process 26840) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

Lab Task 13: Working with Session, TempData, and Cookies

Session Form - Lab Task 13

localhost:5013/Home/SessionForm

Lab Task 13

1. Session Demo — Enter Your Details

Name

John Thomas

Email

thomas@gmail.com

Save to Session

View Session Data

Session Data - Lab Task 13

localhost:5013/Home/SessionDisplay

Lab Task 13

Session Values

Name John Thomas

Email thomas@gmail.com

Back to Form

Clear Session

Refresh this page (F5) — the values stay, because they're in Session, not TempData.

Add Employee - Lab Task 13

localhost:5013/Home/AddEmployee

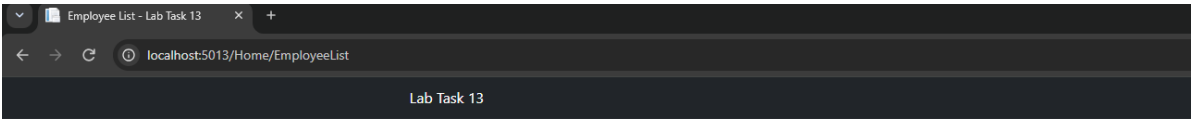
Lab Task 13

2. TempData Demo — Add Employee

Employee Name

John Thomas

Add Employee



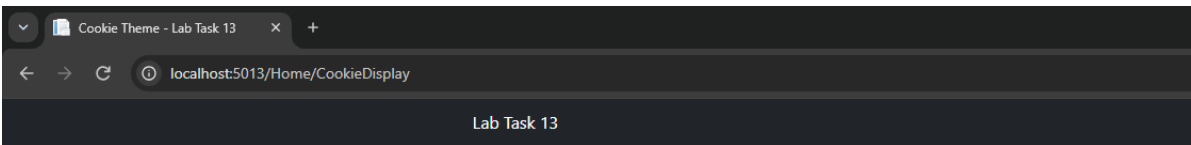
Employee List

Employee 'John Thomas' added successfully!

This message came from TempData. It survived the redirect from AddEmployee, but will NOT survive another request.

Add Another Employee

Reload This Page (simulates a second request)



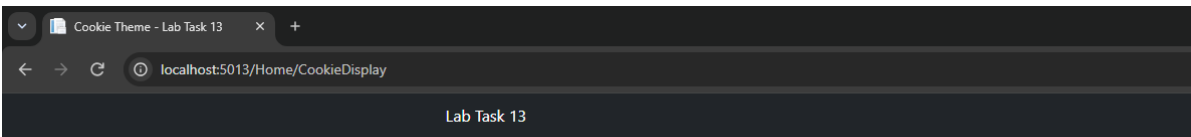
Current Theme: Light

This box's background/text color is set from the ThemePreference cookie.

Change Theme

Delete Cookie

Cookie set with Expires = DateTimeOffset.Now.AddDays(1) — persists 1 day.



Current Theme: Dark

This box's background/text color is set from the ThemePreference cookie.

Change Theme

Delete Cookie

Cookie set with Expires = DateTimeOffset.Now.AddDays(1) — persists 1 day.